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Dynamic Prediction of *Waiting Time Intervals*

Application to Cancer Care Facilities

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Why waiting time matters

Ibrahim, R. (2018)

01

Reduce uncertainty

Patients know what to expect — less cognitive load waiting for news.

02

Increase control

A clear estimate lets patients plan coffee, calls, or returning later.

03

Sense of progress

Seeing time move brings a feeling of forward motion, even during delays.

Two worlds of waiting

Two queueing regimes with very different dynamics

EMERGENCY DEPARTMENT

Arrivals

Walk-ins arrive stochastically, with heterogeneous acuity and limited control over timing.

Queue discipline

APPOINTMENT-BASED CLINIC

Arrivals

Patients are booked into slots and then may be early, late, cancel, or not show at all.

Queue discipline

Takeaway *Emergency waiting is mainly about pooled congestion; appointment waiting is about schedule-state feedback.*

Why waiting is difficult to predict

73

NCI-designated cancer centers
in the United States

A typical patient flow



Process heterogeneity

- Multi-stage (bloodwork, exam, infusion)
- Provider specializations
- Patient-specific treatment plans

Operational uncertainty

- Stochastic arrivals & durations
- Varying service needs & availability
- Downstream capacity feedback

Implication Waiting time is not driven by a single queue — it emerges from interactions across stages, resources, and patient-specific pathways.

Three ways to communicate a prediction

And the trade-offs between them

POINT

42 min

Strength

Simple to state

Limitation

Over-precise in noisy systems; hides operational uncertainty.

INTERVAL

[35, 50] min

Strength

Communicates uncertainty while staying easy to read

Limitation

Need to carefully choose quantiles.

PROBABILISTIC

$p(\text{wait} \leq t)$

Strength

Most information-rich representation

Limitation

Harder to explain, calibrate, and act on in real time.

Our choice *Intervals balance interpretability and uncertainty better than a point estimate or a full distribution.*

Research question

Can we accurately predict waiting times,
in real time, as intervals that update
as the queue unfolds?

REQUIREMENT 01**Dynamic, real-time**

Delivered instantly, continuously updated as queue state changes.

REQUIREMENT 02**Intervals, not points**

Both a lower and upper bound — honest about uncertainty.

Chapter 1

Results & contribution

- Applied a probabilistic model to predict waiting time intervals,
- based on Random Survival Forest
- Optimized quantiles to devise narrower bounds
- Extensive numerical experiments against alternative models,
- including ANN, Quantile regression, CoxPHSurvival, NN-Survival, CTree
- Developed a simulation-based benchmark as a performance upper bound
- Provided a practical solution deployed to an appointment-based
- cancer-care hospital

HEADLINE RESULT

38-minute
average interval width
79% coverage

on held-out test set

Outline

Where we are

- 01 Motivation
- 02 ***Literature Review***
- 03 Predictive Model
- 04 Numerical Experiments
- 05 Implementation
- 06 Conclusion & Next Steps

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Literature review & positioning

P R I O R W O R K

Waiting-time predictions

- Theoretical (Whitt 1999)
- Empirical (Dong et al. 2019)
- Behavioral (Liu et al. 2018)
- Methodological (Meinshausen 2006)

A P P L I E D

ML-based estimation

- Point: Ang et al. (2016),
Kuo et al. (2020)
- Interval / Probabilistic:
Ding et al. (2010), Sun et al.
(2012), Salari et al. (2022),
Arora et al. (2023)

O U R C O N T R I B U T I O N

Shipped end-to-end

- Predictive model for appointment-based systems
- Intervals via customized quantiles (duration vs. accuracy trade-off)
- Practical solution deployed to cancer-care hospital

Outline

Where we are

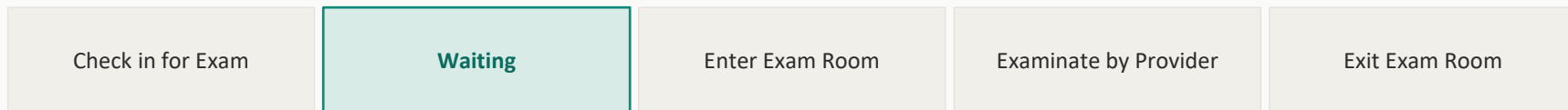
- 01 Motivation
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A typical patient flow

We focus on the exam waiting time



EXAM WAITING TIME



what we predict

Feature set

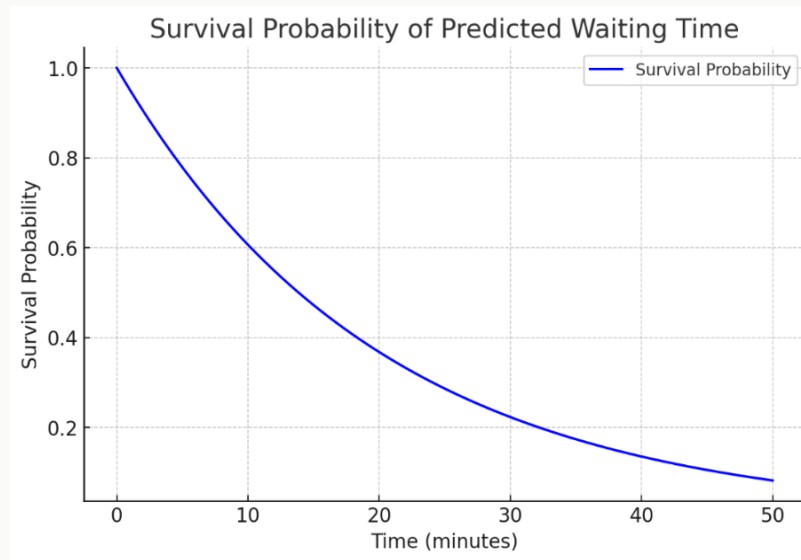
Three groupings capture schedule, real-time, and provider context

Category	Feature	Explanation
Planned schedule	Plan Visit Type	Reason for visit
	Planned time	Planned Exam start time
	Planned position in queue	How many patients are planned to start before this patient for the same provider
	Plan Gap Time	Time from the former planned patient's planned end to planned start
	Lab Before Exam	Whether patient has Lab before Exam
	Infusion Before Exam	Whether patient has Infusion before Exam
Real-time information	Arrival hour	Hour of arrival time
	Arrival to plan start	Time (in minute) from actual arrival to planned time
	Actual position in queue	Number of patients in the front queue for the same provider
	Actual position in queue (consider plan schedule)	Number of patients in the front queue that are planned to start before this query patient for the same provider
	Planned duration of patients in front	Total planned duration of the patients in the front queue and in service minus in service time
	Provider inservice time	Time (in minutes) since provider last started service if provider is now in service
	Provider idle time	Time (in minutes) since provider last finished service if provider is now idle
Provider statistics	Department	Department the provider is affiliated with
	Prob_delay15	Probability of provider being late for the first appointment for more than 15 minutes
	Prob_delay30	Probability of provider being late for the first appointment for more than 30 minutes
	Prob_delay90	Probability of provider being late for the first appointment for more than 90 minutes

Random Survival Forest

A nonparametric ensemble, ideal for time-to-event

- Predicts the survival probability ($1 - \text{CDF}$) of time-to-event
- Supports interval predictions via quantiles
- Supports dynamic predictions using conditional probability



Two key adjustments

To make the forest both faster and tighter

01

Round waiting times

Rounds to 1-minute resolution — reduces tree size and speeds up inference without material accuracy loss.

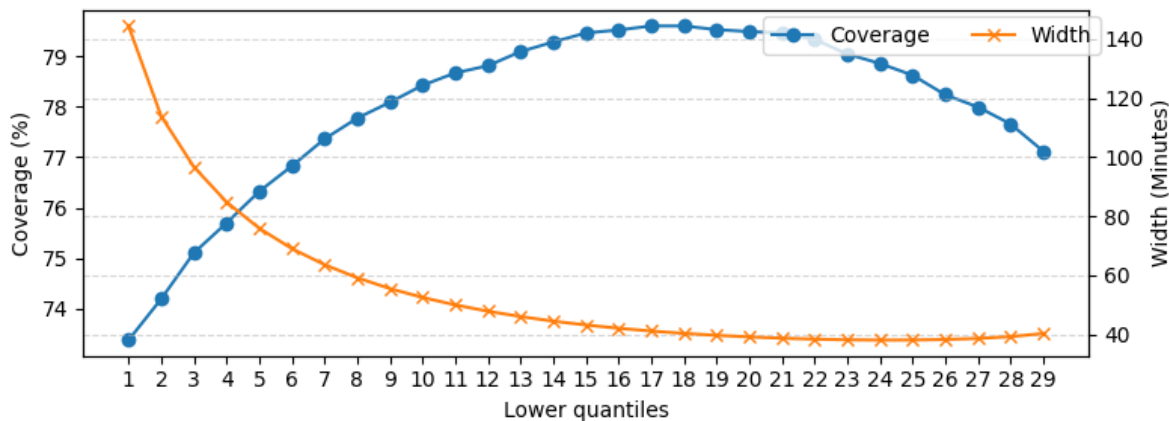
02

Use optimized quantiles

Rather than fixed 15th / 85th bands, pick the quantile pair that gives the narrowest interval at target coverage.

Quantile selection in practice

An example of tightening the interval at a target coverage



TARGET COVERAGE

70%

Fixed (15th, 85th)

42 minutes

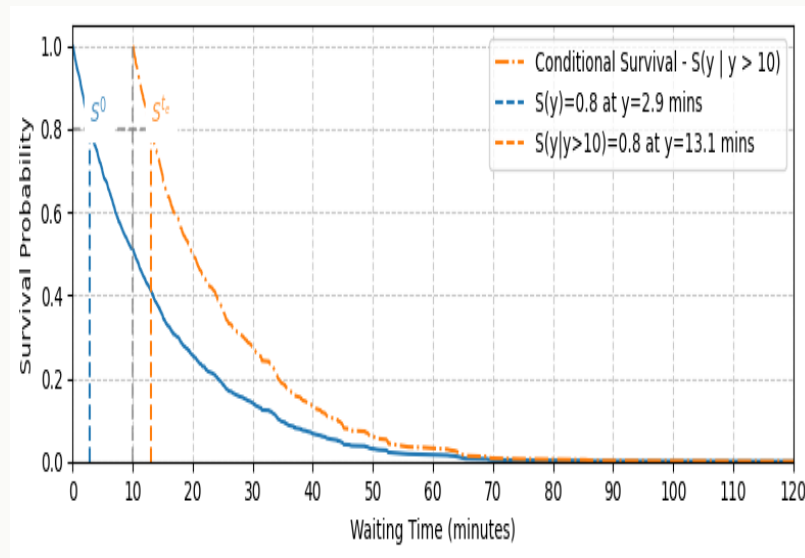
Optimized (26th, 96th)

38 minutes

Dynamic prediction

Updating the prediction as the queue unfolds

- Feature update (e.g., number of waiting patients ahead)
- Conditional distribution (given patient i has already been waiting t minutes)



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Performance measures

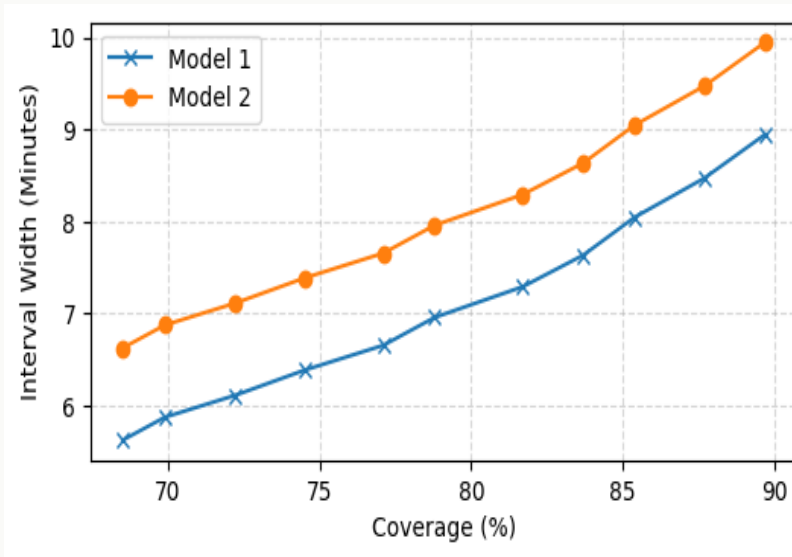
Trading off coverage against interval width

INTERVAL WIDTH

Average interval width

COVERAGE

Probability the realized wait falls inside the interval



Dataset

Two data sources, one unified patient record

Encounter data (planned schedule)

Scheduled start time, duration, provider, reason for visit, etc.

RTLS data (actual timestamps)

Real-time records of each process
— arrival, enter room, exit room.

TRAIN / VAL / TEST SPLIT

Train Oct '22 – Sep '24	301,975
Validation Oct '24 – Dec '24	39,602
Test Jan '25 – Jun '25	91,520

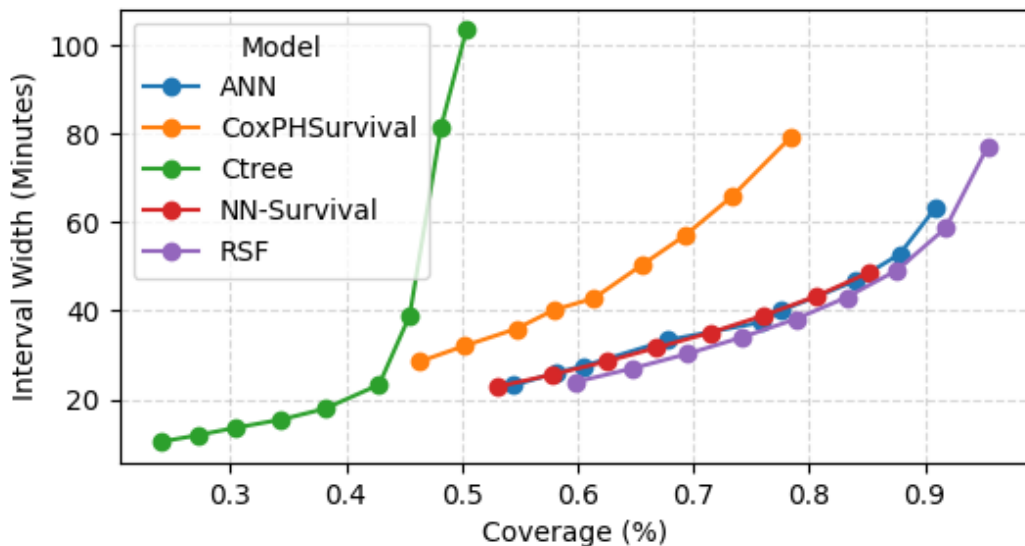
Data example

A single patient row, reconstructed from encounter + RTLS

ColumnName	Explanation	Example	Source
Date	Date of visit	1/2/2024	Action log
Patient	Patient ID	Patient X	Action log
Activity	Type of activity	Exam	Action log
Plan ArrivalTime	Planned patient arrival time (assume punctual arrival)	1/2/24 9:30 AM	Encounter
Plan ServiceStartTime	Planned patient start time	1/2/24 9:30 AM	Encounter
Plan ServiceEndTime	Planned patient end time	1/2/24 10:00 AM	Encounter
Plan EncounterType	Reason for visit	Consultation	Encounter
Actual ArrivalTime	Time when patient checks in for activity	1/2/24 8:55 AM	Action log
Actual ServiceStartTime	Time when patient enter exam room	1/2/24 9:02 AM	Action log
Actual ServiceEndTime	Time when patient exit exam room	1/2/24 9:25 AM	Action log
Plan Resources	Planned provider ID	Provider Y	Encounter
Actual Resources	Actual provider ID	Provider Y	Action log
Plan Location	Scheduled center and floor	Center A, Floor 1	Encounter
Actual Location	Actual center and floor	Center A, Floor 1	Action log
Plan Department	Scheduled department name	Department B	Encounter
Actual Department	Actual department name	Department B	Action log

Comparison with other models

RSF, ANN, and NN-Survival outperform the rest

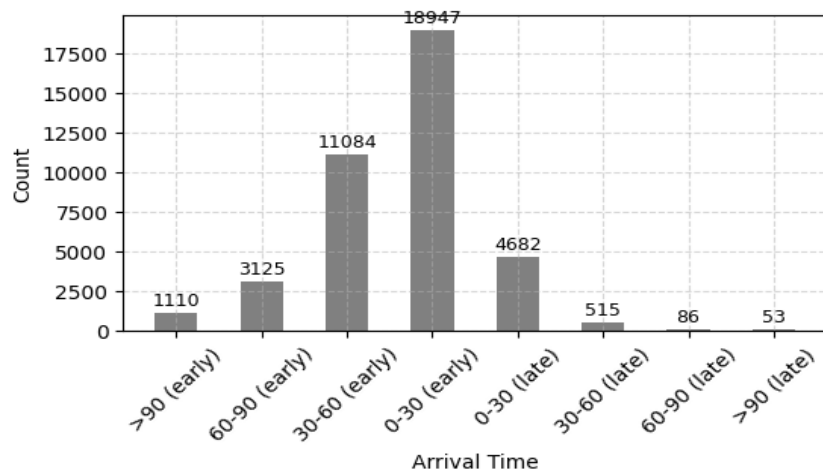
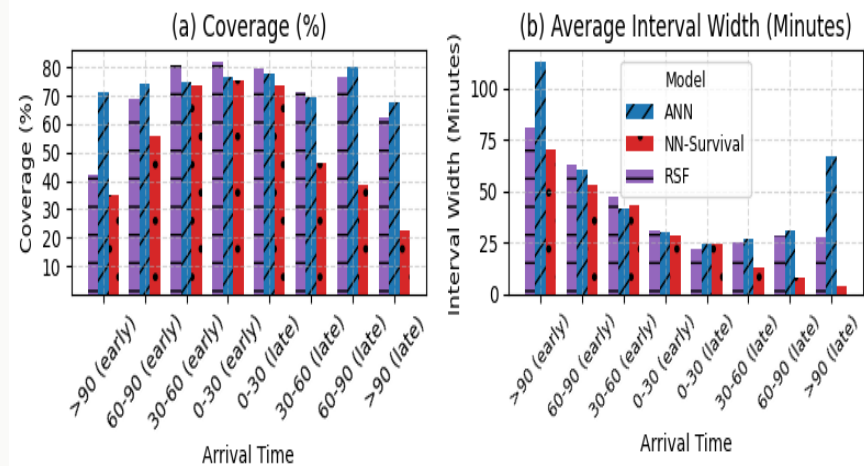


FINDING

ANN, *NN-Survival*, and *RSF* dominate the other two baselines across the coverage–width curve.

Performance by arrival group

Count of encounters vs. actual minus planned arrival



Error analysis

For uncovered observations, how far does the true wait fall outside?

Error = actual wait – lower bound if actual < lower actual wait – upper bound if actual > upper

NEGATIVE ERROR

Upper bound is too high —
interval overestimates waiting.

POSITIVE ERROR

Lower bound is too low —
interval underestimates waiting.

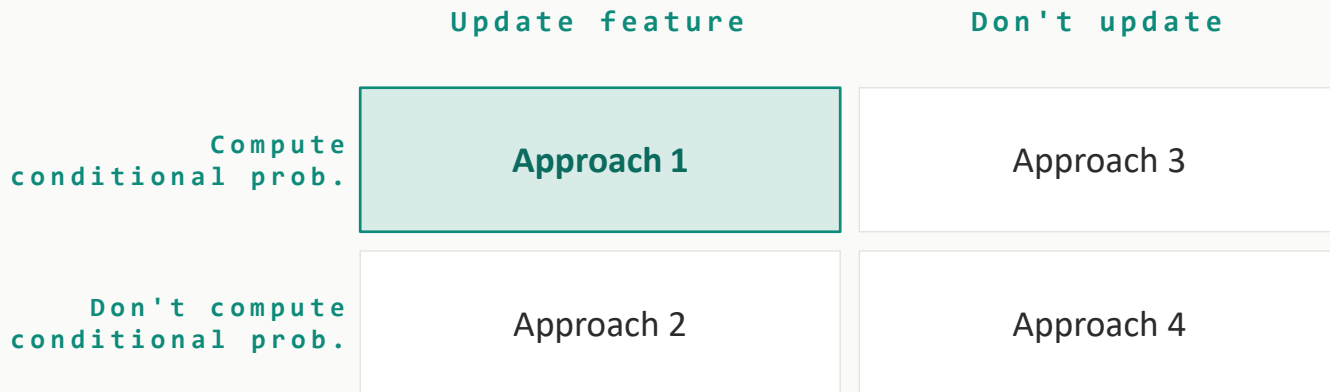
Table 6 Summary statistics of differences for uncovered observations (based on the validation set).

Model	Encounters	Mean	Std	Min	5%	25%	50%	75%	95%	Max
Negative difference										
ANNs	3,380	-4.26	4.98	-45.27	-14.39	-5.53	-2.54	-1.04	-0.19	-0.00
NN-survival	2,737	-3.99	6.64	-120.00	-12.00	-4.00	-2.00	-1.00	-1.00	-1.00
RSF	1,427	-2.76	3.57	-28.50	-9.26	-3.36	-1.58	-0.67	-0.13	-0.02
Positive difference										
ANNs	6,176	15.97	25.82	0.00	0.62	3.94	9.45	18.86	47.86	470.1
NN-survival	8,552	18.36	25.05	1.00	1.00	5.00	11.00	22.00	55.00	432.00
RSF	7,442	17.56	27.47	0.02	0.77	4.05	10.10	20.40	55.76	450.28

Note: negative (positive) difference refer to the actual waiting time minus the lower (upper) bound, in minutes.

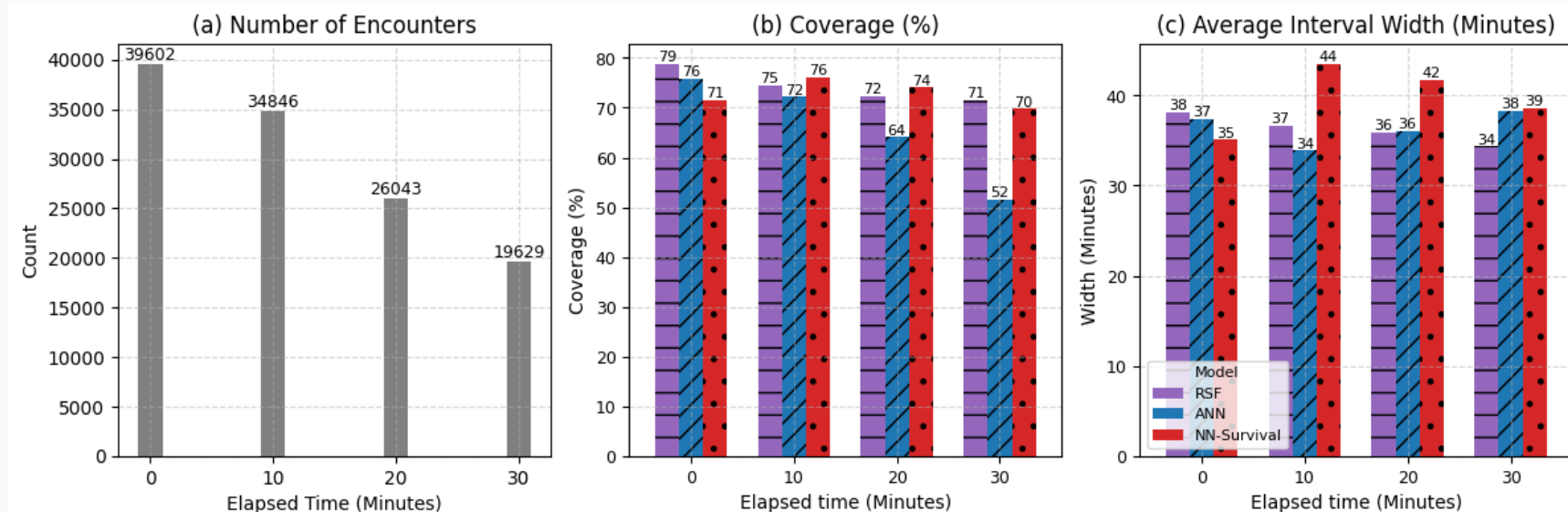
Dynamic prediction: four approaches

Cross feature-update with conditional probability



Comparison at different prediction times

RSF, ANN, and NN-Survival across elapsed-time buckets



Is an ML model worth the effort?

Compare model against two simpler baselines

BASELINE 01

Planned schedule

Use the schedule directly — planned start ± 15 min and planned start ± 20 min.

BASELINE 02

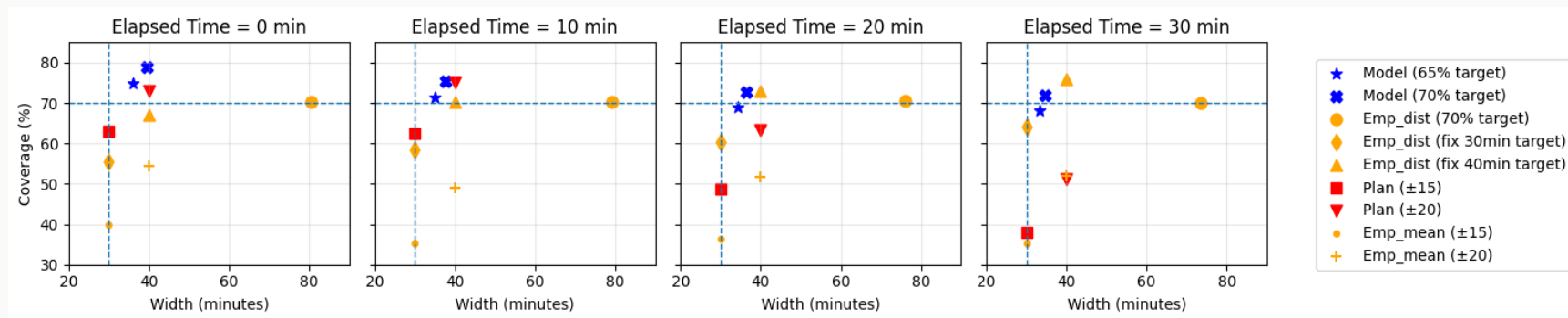
Empirical distribution

Use empirical waiting-time distribution.

- (a) Fix 70% coverage — find narrowest interval.
- (b) Fix 30 / 40 min width — find max coverage.

Comparison across elapsed time

ML model vs. planned schedule vs. empirical distribution



Is a real-time tracking system worth maintaining?

Table 4 Coverage and interval width at different elapsed times under different data inputs (based on the validation set and RSF model).

Elapsed time (t_e)	Without RTLS data, no arrival time		Without RTLS data, use arrival time		With RTLS	
	Coverage (%)	Interval width	Coverage (%)	Interval width	Coverage (%)	Interval width
0 min	73.81	43.97	74.09	43.17	78.85	38.12
10 min	66.33	35.82	67.26	36.24	74.56	36.66
20 min	67.23	35.02	68.52	35.61	72.40	35.90
30 min	68.23	33.21	69.83	33.93	71.44	34.32

INTERPRETATION

Best value at arrival

RTLS is most useful at $t = 0$, when the system knows least about the patient's future progress.

Close to a simpler proxy

Adding only arrival time already gets fairly close to RTLS-level performance.

Operational implication

Predictive model is still valuable even without an RTLS system.

Take away *Real-time tracking helps, but the gain is moderate — not transformative.*

SIMULATION-BASED BENCHMARK

Is our current model *good enough*?

Why are the intervals relatively wide (~ 38 min)?

Two possibilities:

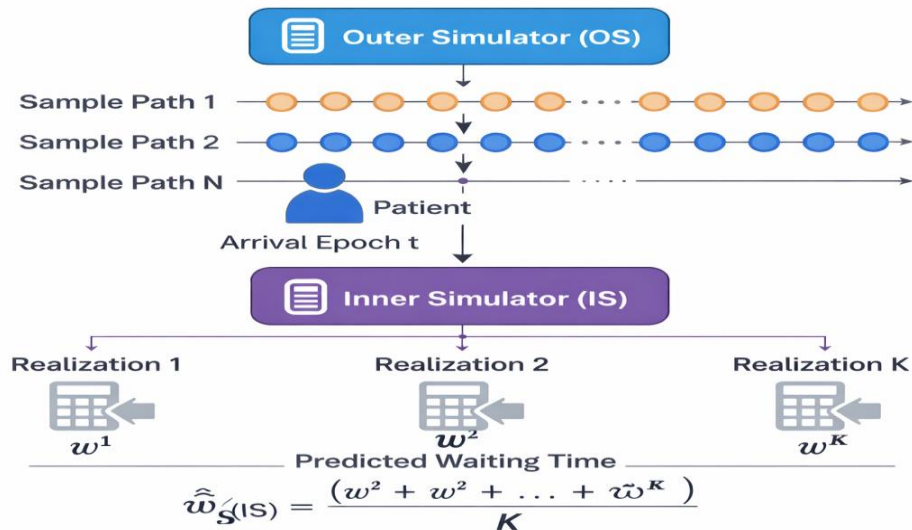
- Data or model limitation
- Irreducible error from stochastic service

APPROACH

Simulate the service system to evaluate the best attainable predictor.

Nested simulator framework

Inner and outer simulators evaluate the best attainable predictor

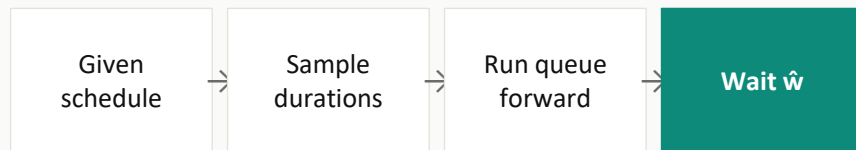


Assumptions & interval construction

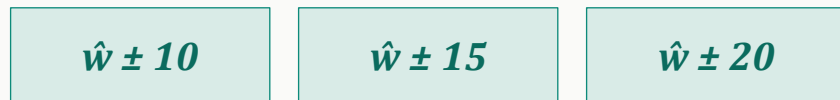
ASSUMPTIONS

- Consider a single provider's session
- Provider-level schedule given
- Arrivals are punctual
- Service durations from fitted provider-level distribution
- Patients served strictly in scheduled sequence
- No provider gap time

INTERVAL CONSTRUCTION

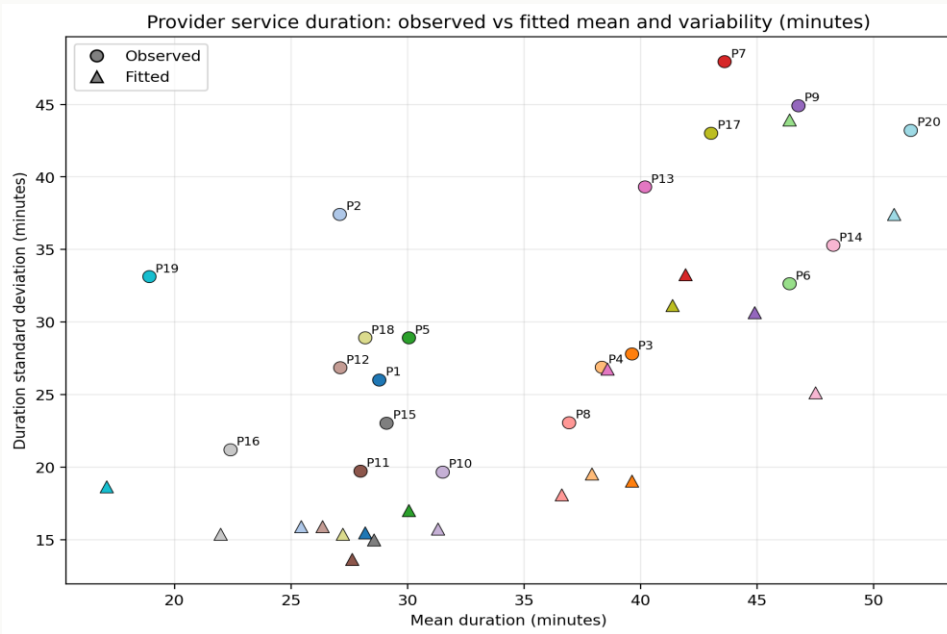


POSSIBLE INTERVALS



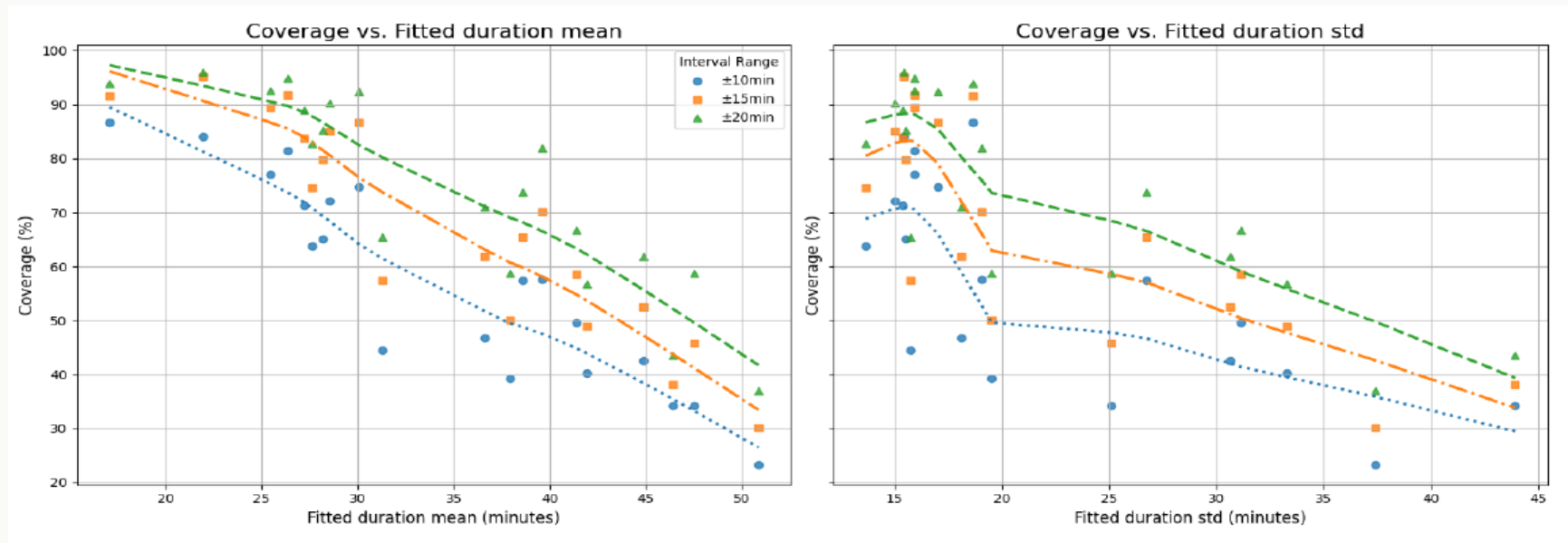
Provider parameters

Twenty providers, with fitted duration distributions

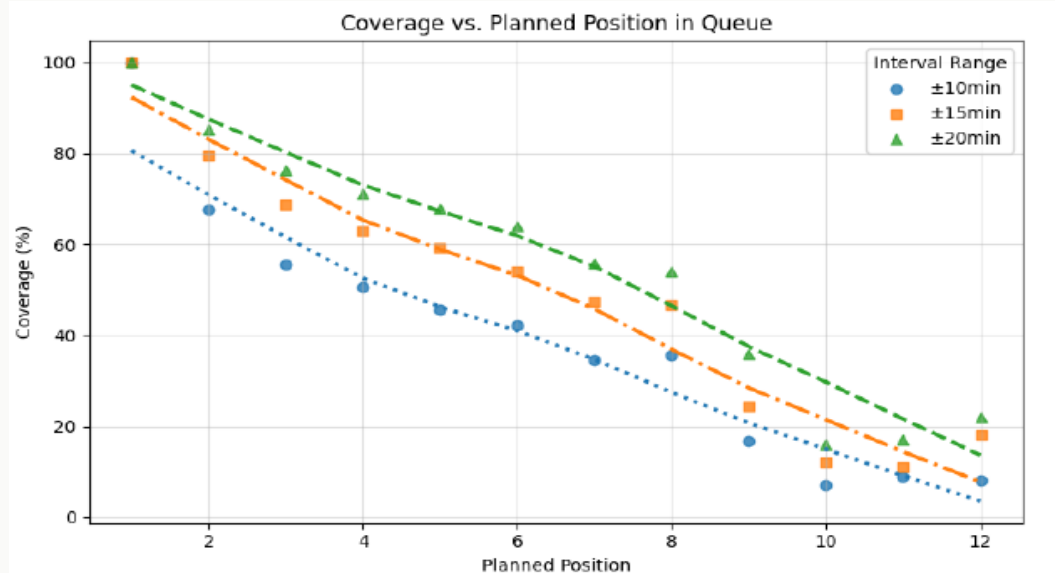


Coverage vs. fitted duration

Results across provider-level fits — mean and std



Coverage vs. planned position in queue



TAKEAWAY

Error accumulates with the queue.

A refined assumption

Letting the first patient's start time depend on arrival

REVISED ASSUMPTIONS

- Consider a single provider's session
- Provider-level schedule given
- ~~• Arrivals are punctual~~
- Service durations from fitted provider-level distribution
- Patients served strictly in scheduled sequence
- ~~• No provider gap time~~

NEW DEFINITION

**First patient's
start time**

= arrival time + provider gap time

Start-time prediction for patient i

Patient $i-1$ in service; no other patient before patient i

Patient i 's start time = (remaining service time of patient $i - 1$) + provider gap time

Method	Coverage	Width	MAE
Remaining dur + gap (predicted start ± 15 min)	62.7 %	30 min	14.6 min
Remaining dur + gap (predicted start ± 20 min)	72.1 %	40 min	
RSF interval	79.9 %	38 min	13.6 min

Four experiments, four questions

01 Performance of ML models

- Compare several ML models
- Evaluate at arrival, +10, +20, +30 min
- Include error distance for uncovered waits

02 Value of the ML model

- Compare best ML with planned schedule
- Compare best ML with empirical distribution
- Whether learning beats simple operational rules

03 Value of the RTLS system

- With vs. without real-time inputs
- Quantify marginal value of RTLS-like data
- Whether simpler feature sets already get close

04 Simulation upper bound

- Benchmark against best attainable simulation
- Separate limits from irreducible uncertainty
- Interpret the remaining gap to the bound

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Predictive tool workflow

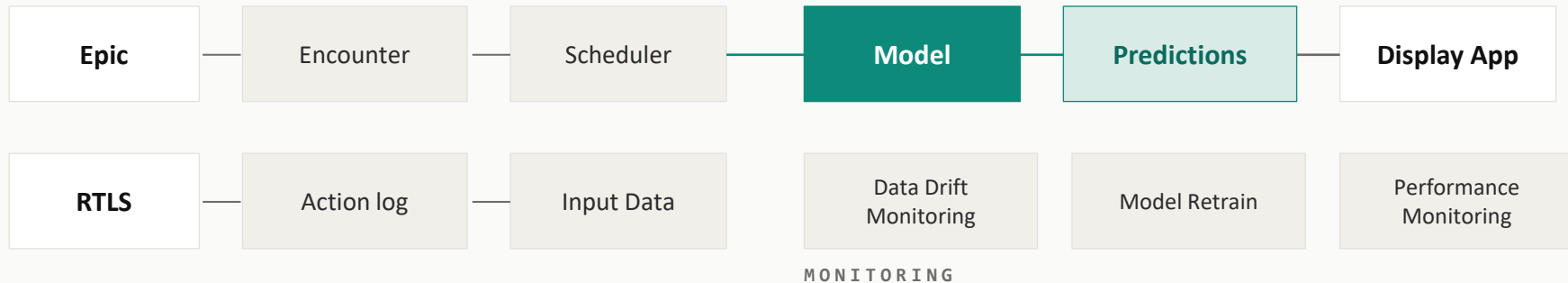
Three lanes: data management, modeling, and software engineering

DATA MANAGEMENT

SCHEDULING

MODELING

DEPLOYMENT



Display application

Updates every 5 minutes, one floor at a time, shown to clinical assistants

Select a Clinic
DF YAWKEY 7

Model v1.0 | Run #1591
Refreshed 6 minutes ago

Patient Name	DFCI MRN	Appointment Date	Expected Wait Time	Upper Bound	Lower Bound	Provider Name	Department	Visit Type
		08/11/2025 15:30 PM	50 minutes	74 minutes	5 minutes		DF LYMPHOMA	ESTABLISHED PATIENT
		08/11/2025 15:00 PM	45 minutes	57 minutes	13 minutes		DF LYMPHOMA	TEACHING / STUDY
		08/11/2025 15:00 PM	26 minutes	34 minutes	6 minutes		DF MULTIPLE MYELOMA	ESTABLISHED PATIENT ON TX
		08/11/2025 14:30 PM	27 minutes	37 minutes	4 minutes		DF MULTIPLE MYELOMA	ESTABLISHED PATIENT ON TX
		08/11/2025 14:30 PM	42 minutes	50 minutes	4 minutes		DF MULTIPLE MYELOMA	ESTABLISHED PATIENT
		08/11/2025 15:30 PM	61 minutes	75 minutes	26 minutes		DF MULTIPLE MYELOMA	ESTABLISHED PATIENT ON TX
		08/11/2025 11:00 AM	170 minutes	311 minutes	30 minutes		DF LYMPHOMA	CONSULT
		08/11/2025 14:30 PM	18 minutes	25 minutes	3 minutes		DF MULTIPLE MYELOMA	ESTABLISHED PATIENT
		08/11/2025 14:30 PM	18 minutes	25 minutes	3 minutes		DF LYMPHOMA	ESTABLISHED PATIENT

Select a Clinician

Patient Name	DFCI MRN	Appointment Date	Expected Wait Time	Upper Bound	Lower Bound
	1162857	08/11/2025 16:00 PM	46 minutes	65 minutes	8 minutes
	1319552	08/11/2025 15:00 PM	29 minutes	33 minutes	3 minutes

Display application — live results

Coverage and width by elapsed time since arrival

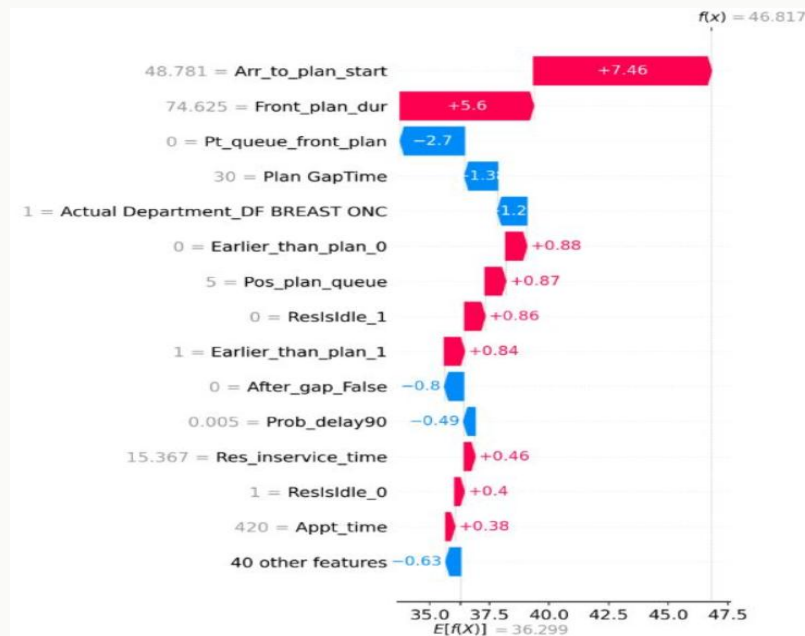
Elapsed time	Predictions	Coverage	Width
minutes from arrival to prediction	count in bucket	(%)	(minutes)
0 – 5	1,563	79.91	40.69
5 – 10	2,176	79.92	40.45
10 – 15	1,930	79.53	40.00
15 – 20	1,698	78.09	39.34
20 – 25	1,499	77.85	38.28
25 – 30	1,312	76.75	37.35
30 +	6,656	73.35	35.15

Model interpretability via SHAP

Attributing each prediction's deviation from the reference to features

$$y_i - \bar{y} = \sum \varphi_k(x_k)$$

For each visit, SHAP attributes the deviation of the predicted expected wait from a reference prediction to individual input features.



RSF-to-RF approximation

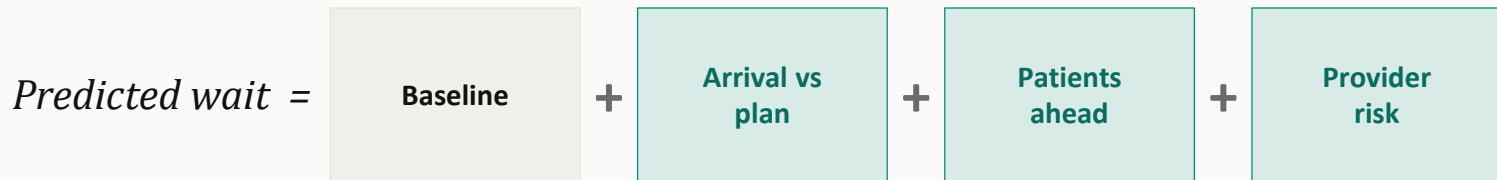
Convert each survival tree into a regression-style tree for SHAP

$$\mu^{RF}(y) = \int \hat{S}(y | x) dy$$

- Keeps the same predicted mean waiting time as RSF
- Dramatically reduces explanation runtime and memory cost
- Makes SHAP practical for large-scale analysis

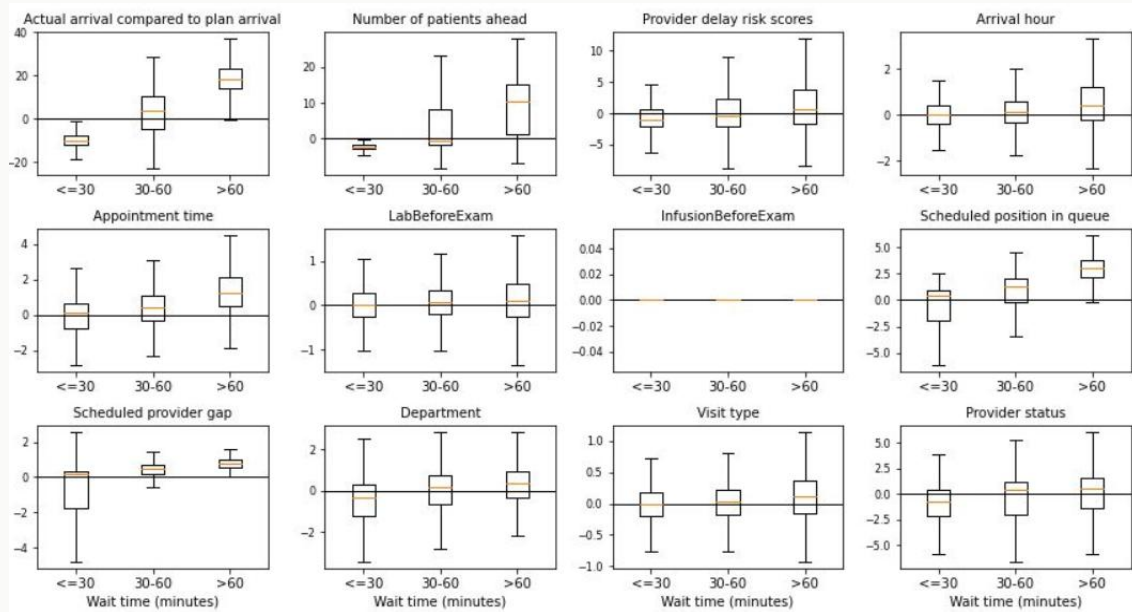
Feature grouping

Operational groupings read faster than raw feature names



Operational groups make explanations faster to read than dozens of raw feature names.

Feature importance across wait buckets



MAIN DRIVERS OF LONG WAITS

- Actual vs. planned arrival has
- the widest SHAP range — the
- dominant driver.
- Number of patients ahead is the
- clearest real-time congestion signal.
- Provider delay-risk scores matter,
- especially for longer waits.

Prediction result interpretation

A per-visit explanation combining prediction + natural-language drivers

Explanation	Expectation	Date	Patient	Lower Bound	Upper Bound	Plan ServiceStartTime	Actual ArrivalTime	Actual ServiceStartTime
Expected wait time: 42.6 minutes Average wait time: 35.9 minutes Delay reasons: 1 patients waiting ahead (+6.4 minutes) Provider's delay risk scores (0 lowest, 1 highest): 0.22 (-2.2 minutes) Arrived 30 minutes early (+1.6 minutes) Scheduled appointment time is: 09:00 (-1.5 minutes) Provider has been idle for: 0.0 minutes (+1.2 minutes)	42.13	2025-03-17	85FACF8A-E3	21	54	2025-03-17 09:00:00	2025-03-17 08:29:33	2025-03-17 09:13:01
Expected wait time: 48.8 minutes Average wait time: 35.9 minutes Delay reasons: Arrived 31 minutes early (+4.2 minutes) Provider's delay risk scores (0 lowest, 1 highest): 0.46 (+2.4 minutes) Scheduled appointment time is: 15:30 (+1.6 minutes) Time gap between scheduled appointments is: 0 minutes (+1.4 minutes) Scheduled position in queue is: 4 (+1.4 minutes)	48.25	2025-01-27	79968B5B-117	13	63	2025-01-27 15:30:00	2025-01-27 14:59:15	2025-01-27 15:43:49
Expected wait time: 54.1 minutes Average wait time: 35.9 minutes Delay reasons: 1 patients waiting ahead (+13.8 minutes) Provider's delay risk scores (0 lowest, 1 highest): 0.51 (+5.0 minutes) Arrived 26 minutes early (-3.5 minutes) Scheduled position in queue is: 5 (+2.2 minutes) Arrival hour: 14 (-1.2 minutes)	53.60	2025-03-05	DF86E007-AE	20	69	2025-03-05 14:30:00	2025-03-05 14:04:26	2025-03-05 15:18:07
Expected wait time: 46.7 minutes Average wait time: 35.9 minutes Delay reasons: 1 patients waiting ahead (+7.3 minutes) Time gap between scheduled appointments is: 30 minutes (-3.8 minute) Arrived 34 minutes early (+2.3 minutes) Scheduled position in queue is: 2 (+1.4 minutes) Provider has been with other patients for: 54.34 minutes (+1.0 minutes)	46.15	2025-06-27	A3E2E25E-1D	17	60	2025-06-27 11:30:00	2025-06-27 10:56:19	2025-06-27 12:00:01

However — we don't show SHAP to patients

STAKEHOLDER CONCERN

Attributing delays to specific operational factors may be perceived as poor service quality — and could negatively affect patient satisfaction.

CURRENT ROLE OF SHAP

Internal qualitative sanity check. We validate whether the model's behavior matches clinical-staff intuition.

Lessons learned

Cross-team coordination

Deployment required tight collaboration across clinical assistants, data management, model developers, and software engineers — clear workflow definitions and documentation were essential.

Processing & traceability

Real-time systems need freshness checks, output monitoring, and error alerts. Version control, change logs, and rollback paths are critical for safe operations.

Time & financial investment

A production tool takes substantial time beyond model building — stakeholder meetings, workflow alignment, and cloud operations all add real cost.

Adoption changed the product

Users often preferred outputs we hadn't originally designed for — a simple point estimate or a threshold probability instead of an interval.

A key reflection — flexible outputs matter

ORIGINAL DESIGN GOAL

Interval output

[lower bound , upper bound]

Good for communicating uncertainty.

OBSERVED ADOPTION PATTERN

Point estimate

Expected wait time

Threshold probability

$P(\text{wait} > 30 \text{ min})$

Interval output

Lower + upper bounds

- Flexible output formats are a product requirement, not just a modeling choice.
- Future evaluation should compare not only interval quality but also derived point estimates and threshold-based decisions.
- A full predictive distribution is valuable because it supports multiple presentation formats.

Dynamic prediction, *deployed*.

- Developed a Random Survival Forest model to predict waiting-time intervals
- Extensive numerical experiments benchmarked against alternative approaches
- Provided a simulation-based achievable upper bound for the best attainable predictor
- Explored model explainability with SHAP and operational feature grouping
- Piloted and deployed real-time prediction in a cancer-care hospital

Thank you. Questions?

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